

Embedded System Workshop (EC3.202)

Introductory Class

Instructors: Sachin Chaudhari, Aftab Hussain, Deepak G.

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INTERNATIONAL INSTITUTE OF
INFORMATION TECHNOLOGY

HYDERABAD

Background: Sachin Chaudhari (SPCRC and SCRC)

- **Academics**

- ECE Coordinator, IIITH (2022-present)
- Associate Professor, Head, SPCRC and SCRC, IIIT Hyderabad (2021-present)
- Assistant Professor, SPCRC and SCRC, IIIT Hyderabad (2015-2021)
- Postdoc, Aalto University, Finland (2013-2014)
- PhD, Aalto University, Finland (2007-2012); M.E., IISc Bangalore, India (2002-2004); B.E., VNIT, Nagpur (1998-2002)

- **Industry**

- Senior Wireless Communication Engineer, Esqube Communications, Bangalore (an IISc start-up) (2004-2007)

- **Research Interests:**

- Signal Processing and Machine Learning for Wireless Communication:
- IoT for Smart Cities, 5G and Beyond, Cognitive Radios.

- **Research Projects on IoT**

- **DST and PRIF:** IoT Enabled Smart Cities: Pollution Health and Governance
- **CoE on IoT for Smart Cities:** Coordinator
- **India's First *Living Lab* for Smart City Research**
- **IoT for Education:** Remote Labs
- Several IoT theme projects: Air Pollution, Water Monitoring, Bridge health monitoring, Digital Stethoscope, Garbage Vulnerable Points, Remote Labs
- **Outreach:** College Research Affiliate Program and Talent Sprint course on IoT

Background: Deepak Gangadharan (CSG)

- **Academics**

- Asst Prof., IIIT Hyderabad (2019-present)
- PostDoc Researcher: UPenn (2015-2019), University of Erlangen Nuremberg (2013-2015), Technical Univ. Of Denmark (2012-2013)
- PhD (CS), National University of Singapore (2007-2012)
- BTech (ECE), College of Engineering Trivandrum (1998-2002)

- **Industry**

Design/Senior Design Engineer, Conexant Systems Inc. (2002-2005)

- **Research Interests:**

Cyber-Physical Systems, Real-Time Scheduling, Intelligent Transportation Systems, Edge Computing

- **Research Projects**

- ITS: Traffic flow prediction and control, Vehicular Edge Computing, Digital Twin
- UAV security
- CPS: Real-Time Scheduling for Safety Critical Systems

Background: Aftab Hussain (CVEST)

- **Academics**

- Asst Prof., IIIT Hyderabad (2018-present)
- Postdoc Fellow, Harvard University (2016-2018)
- MS+PhD, KAUST, Saudi Arabia (2011-2016)
- B.Tech, IIT Roorkee (2005-2009)

- **Industry**

- Design Engineer, Analog Devices India (2010-2011)

- **Research Interests:** Flexible electronics, sensor systems, smart cities, IoT

- **Research Projects**

- Flexible pressure sensors based medical devices
- Magnetic actuators based soft robotics
- Sensor/actuator integration
- WiSUN communication for smart city applications

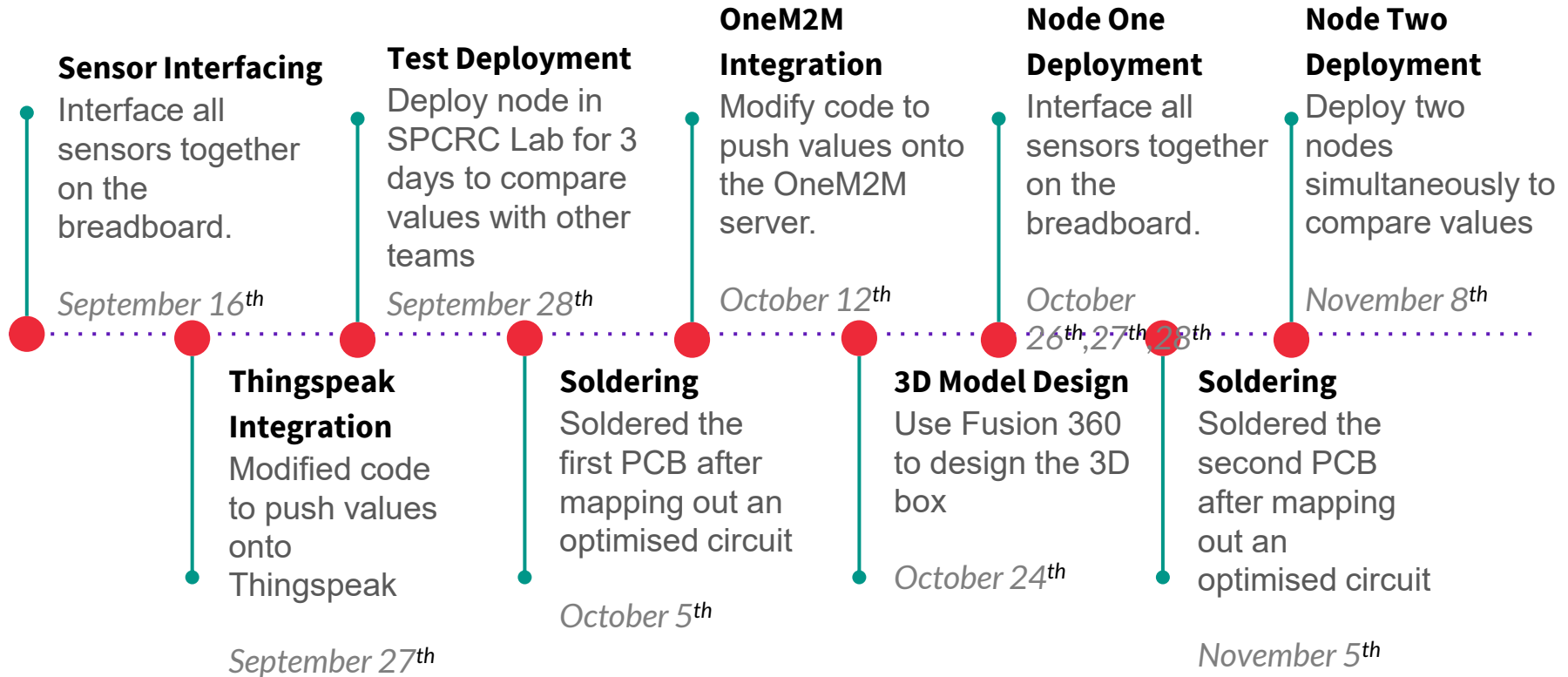
Outline

- Course Intro
 - Embedded System Workshop (Previous years)
 - Projects this time
- Course Administration
 - Resources
 - Evaluation

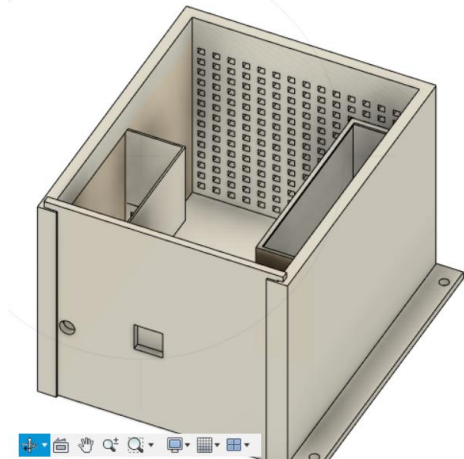
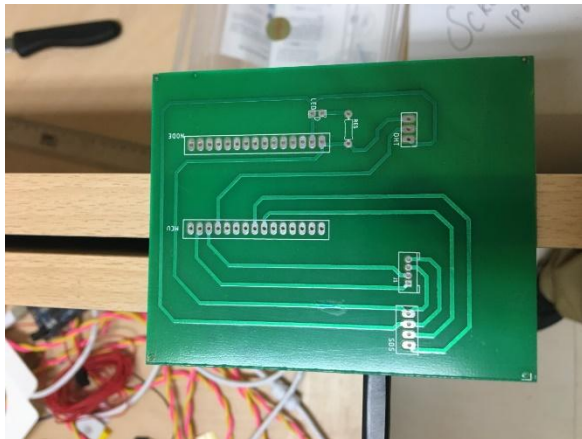
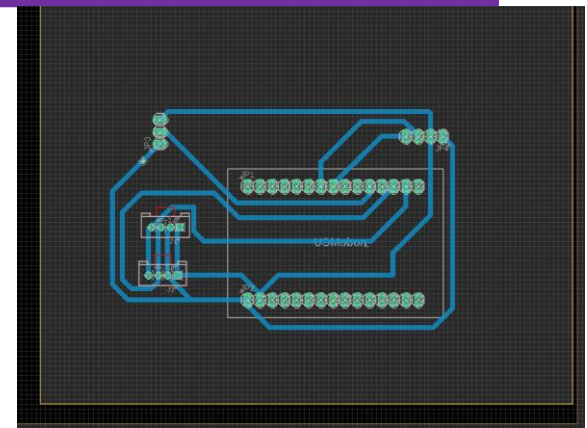
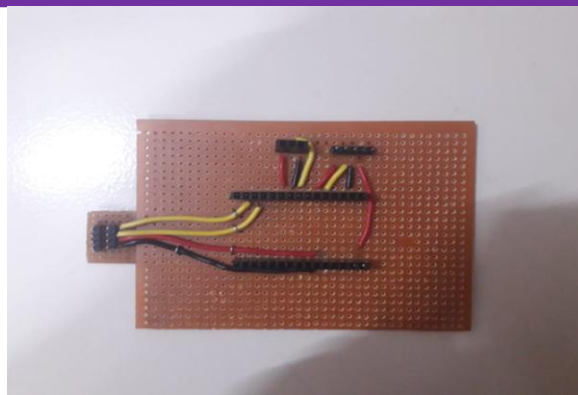
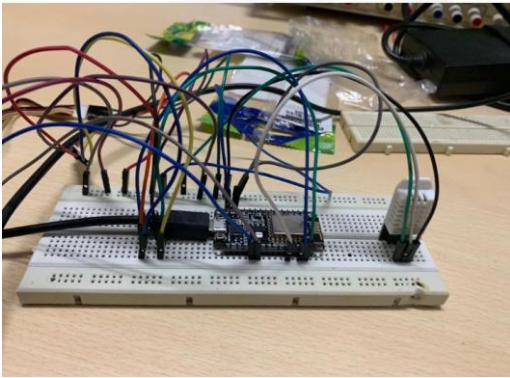
ESW: (M2019-M2021)

- First phase of Smart City Living Lab done by ESW course
 - Coordinators: Vishal Garg, Sachin Chaudhari, Aftab Hussain
 - Hardware provided by institute
 - Topics
 - Air pollution
 - Building energy efficiency
 - Water quality and Quantity
 - Smart classrooms
 - Streetlights
 - And Many more
 - Youtube: <https://www.youtube.com/watch?v=RcxKJ54GH6Q>

ESW: *Different Aspects Involved*



ESW: *Different Phases*



ESW: Some Photos from Deployment

- Indoor: Faculty Quarters – I and II



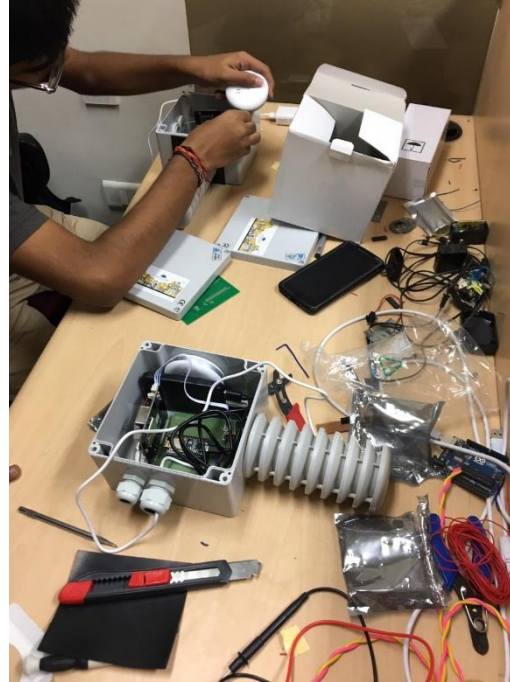
- Indoor: Mess



ESW: Outdoor Deployment

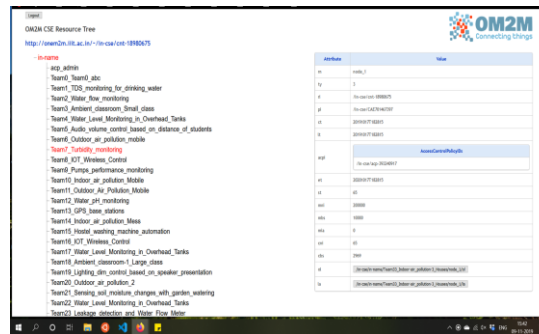
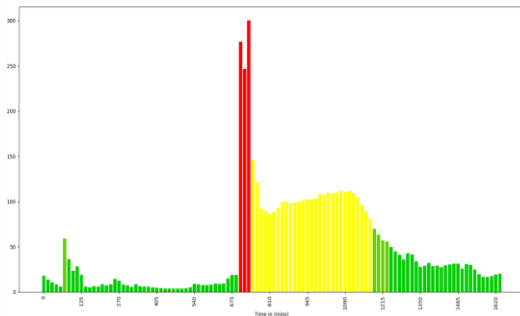
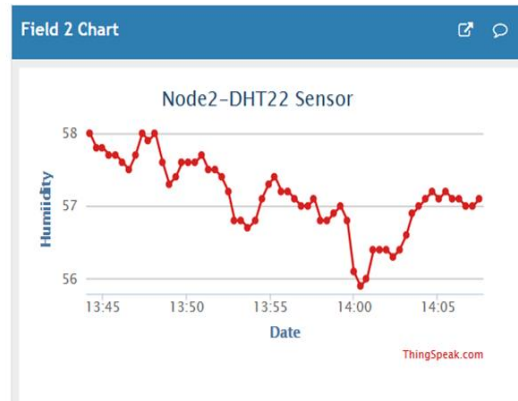
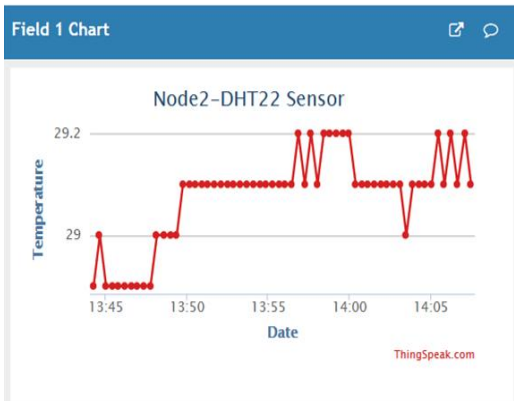
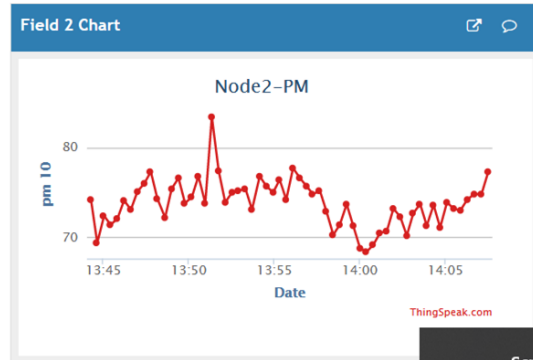
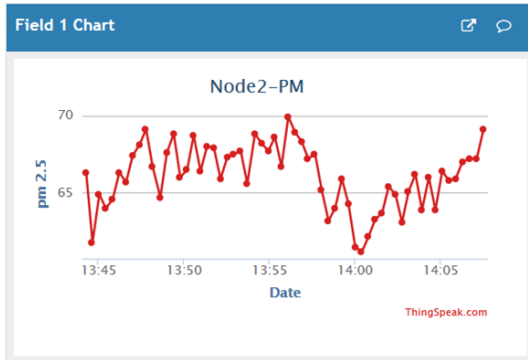


ESW: *Students at Work*



ESW: Sample Data

- ThingSpeak data -



- We have our own Twitter channel !!



oneM2M

ESW 2022-2025

- Publications

- Inesh Dheer, Shreyas Mehta, Srikar Somanchi, Alan Nelson and Abhishek Srivastava, “Pre-Deployment Calibration Framework for Low-Cost Gas Sensors: An Adaptive Environmental Parameter Model,” IEEE Sensors Letters, June 2025.
- Rathan et. al. “Edge-Based Object Classification Using mmWave Radar on Qualcomm Innovators Development Kit” ICEdge 2025

- Introduction of Qualcomm Innovation Development Kits (QIDK) projects in Monsoon 2024 in collaboration with Qualcomm
- Continued in Monsoon 2025

ESW 2022-2025: QIDK projects

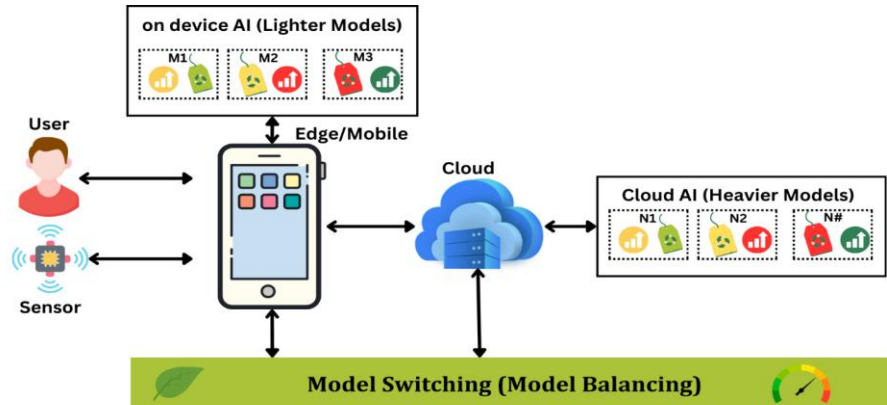
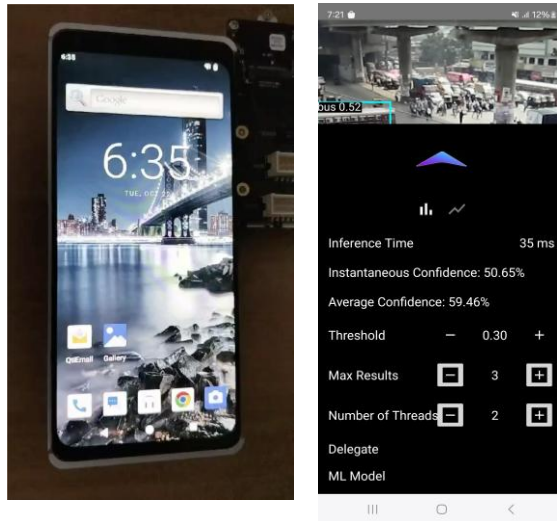
- Several interesting projects
 - Edge ML balancer, AQI estimation using image, pose estimation, waste segregation, Object classification using mmWave radar
 - Few publications
 - SPARQ invitation for 2025 and 2026



SPARQ 2026



Model Switching and Crowd Monitoring



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IIIT-Hyderabad demonstrates ML model-switching approach on smart phones for real-time traffic monitoring

Telangana

M Srinivas

27 February 2025 2:55 PM

Second year CSE students of IIIT-Hyderabad unveiled ML model



A team of 2nd year CSE students of IIIT-Hyderabad has successfully demonstrated a dynamic ML model-switching approach on smart phones for real-time traffic monitoring. (Photo: By Arrangement)

EdgeMLBalancer: A Self-Adaptive Approach for Dynamic Model Switching on Resource-Constrained Edge Devices

Akhila Matathammal*, Kriti Gupta[†], Larissa Lavanya[†],
Ananya Vishal Halgatti[†], Priyanshi Gupta[†], Karthik Vaidhyathan*

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Small Models, Big Tasks: An Exploratory Empirical Study on Small Language Models for Function Calling

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Accepted Papers

1. Murthy, Aryama V., Yashas N. Kotre, Prathmesh Sharma, Pragya Mishra, Sanjith Ganapathi, and Priyesh Shukla. "Phase Matters: Characterizing Heterogeneous Vision-Language Inference on a Mobile SoC." In *Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services Workshops*, pp. 322-327. 2026 ([ACM Mobisys 2026](#))
2. Murthy, Aryama V., Yashas N. Kotre, Prathmesh Sharma, Pragya Mishra, Sushim Shrivastava, and Priyesh Shukla. "TUNE: State-Aware Model-Knob Scheduling Across Edge LLM/VLM Deployments." ([IEEE ICEdge 2026](#))
3. Sri Rama Rathan Reddy Koluguri, Akshat Puneet and Abhishek Srivastava, "Edge-Based Object Classification Using mmWave Radar on Qualcomm Innovators Development Kit," IEEE International Conference on Intelligent Computing and Systems at the Edge (ICEdge), India, Dec 2025. ([Best Student Paper Award](#))
4. Sri Rama Rathan Reddy Koluguri and Akshat Puneet won the Best Student Paper Award at IEEE International Conference on Intelligent Computing and Systems at the [Edge \(ICEDGE\) \(2025\)](#)
5. Sri Rama Rathan Reddy Koluguri, Ravi Theja Kolluru and Hrishikesh Gawas won the AIoT Innovator Award at IEEE International Conference on Intelligent Computing and Systems at the Edge ([ICEDGE](#)) ([2025](#))

Submitted Papers

- Rahul Kumar, Ponnepureddy Srujan, **Sachin Chaudhari**, Saikiran Bulusu, “Segmentation Based IoT System for Garbage Vulnerable Point Monitoring on Edge Devices” submitted to IEEE WF-IoT 2026
- L. Chintada, **S. Chaudhari**, A. Vатtem, “City-Scale Garbage Vulnerable Point Monitoring Using Panoramic Vision and Edge AI” submitted to IEEE WF-IoT 2026

Projects are open-ended!

- Few things which can be common across **IoT projects**:
 - **Soldering, PCB, 3D Enclosure**
 - **Basic dashboard or App**
 - **Data processing / Alerts**
 - **Cloud interfacing**
 - RTC, Power management (putting device to sleep), Security
 - Data transmission and storage reduction, Managing communication outage
- Students are encouraged to test, deploy, and productize their projects
- Same is true for **QIDK projects**
 - Can compare with other computing platforms (QIDK 2.0, Raspberry Pi) if available
 - Try out different models
 - Can explore more possibilities in discussion with faculty and Qualcomm team

IP policy

- Faculty are giving part of their open research problems
 - They will continue doing research on this topic
 - IP and publication rights belongs to the faculty and team
 - If you do sufficient contribution, you can be part of their papers/patent
 - Significant research work needed even after basic PoC
 - Faculty's decision

Projects for ESW based on two themes:

- * IoT**
- * QIDK**

(Tentative themes are provided)

IoT-based projects

Project 1: Multi-Agent Smart Bricks: A Sensor-Based Modular Interactive System

Faculty: Kavita Vemuri

Mentor: Ranapratap

TA: Adeeb

Inspiration :<https://www.lego.com/en-in/smart-play>

Design a set of electronic bricks that communicate with neighbouring bricks and change their behaviour depending on how they are assembled. The first step can be communicate to count presence within a range (for the suggested use case scenario).

Each brick is an autonomous agent that:

- Detects neighboring bricks
- Exchanges information (it can be a code for an action)
- Makes local decisions (sense the surroundings and make decisions)
- Contributes to the overall system behavior

Instead of one controller controlling everything, every brick has its own microcontroller.

Futuristic use case: surgical table where all the instruments communicate with each other to ensure every instrument is accounted for as per requirement.

Project 2: Gait Mat

Faculty: Kavita Vemuri

Mentor: RanaPrathap (from my lab)

TA: Adeeb

Problem Statement: Develop a portable gait mat that measures walking patterns and automatically detects abnormalities associated with conditions such as Parkinson's disease, stroke, diabetic neuropathy, or fall risk.

System to measure & output:

- Step length
- Stride length
- Step width
- Walking speed
- Cadence (steps/min)
- Stance time
- Center of pressure (CoP)
- Foot progression angle
- Left-right symmetry
- Balance during walking

Project 3: Performance Analysis of Wi-SUN Network

- **Faculty Advisor:** Prof. Aftab M. Hussain
- **TA:** Kartik Gharde
- **Abstract:** Wi-SUN (Wireless Smart Utility Network) is an IPv6-based wireless mesh networking technology designed for large-scale smart city and utility applications. IIIT Hyderabad has deployed a campus-wide Wi-SUN network. This project aims to perform a comprehensive performance analysis of the deployed network by evaluating key metrics such as communication range, node connectivity and disconnectivity, network formation time, latency, packet delivery ratio, routing stability, Received Signal Level (RSL), and network recovery under different deployment conditions, etc. The study will identify factors affecting network reliability and provide recommendations for improving the scalability, robustness, and overall performance of large-scale Wi-SUN deployments.
- **Hardware & Software**
 - **Hardware:** Raspberry Pi 5, Wi-SUN Starter Kit, EFR32FG28 Wi-SUN Radio Board, customized PCB
 - **Software:** Simplicity Studio v6, Wi-SUN SDK,

Project 4: Battery-Free, BLE-Enabled Wearable Device for Multi-Floor Indoor Emergency Alert Tracking

•**Faculty Advisor:** Prof. Aftab M. Hussain

•**TA:** Ruthwik M

•**Abstract:** Emergency alert tracking and response is a critical requirement in care facilities, educational institutions, and specialized workplaces. The aim of this project is to replicate and scale up a multi-floor, indoor emergency alert tracking system developed by our research group (corresponding preprint will be shared to the selected students). The system uses battery-free wearable devices that send alerts over Bluetooth Low Energy (BLE). Each wearable is powered by an energy harvesting pushbutton switch that utilizes the mechanical energy of a switch-press to generate BLE advertisements. The Received Signal Strength Index (RSSI) values of these advertisements, captured by static receivers, are used for indoor localization of the devices.

•**Tentative Timeline**

•Week 1: Comprehensive understanding of the existing implementation

•Week 2-3: Development of (a) scalable receiver prototype with custom enclosure (b) final wearable prototype with decal

•Week 4-5: Deployment and data collection in Himalaya; dashboard implementation

•Week 6-7: Transition to a real-time system with dashboard integration

•Week 8-9: Analysis of Himalaya data and external data collection

•Week 10: Conclusion and evaluation

•**Hardware:** PTM216B Switches, Arduino Nano 33 IoTs, Micro-USB cables, USB Power Adapters

Project 5: Smart Farming

- Faculty: Zia Abbas
- TA: Sai Poojith
- In this project, propose an IOT system to sense VOCs for accurately predicting onset of pathogen attack on the plant. This will help to more precisely control the spread of the pathogen by using minimal amount of pesticide on the target area. A network on VOC sensor, temperature, humidity (soil), light, O₂/CO₂ can be deployed on the farm. In the project, these above-mentioned sensor will be interfaced and deployed on an experimental farm (plants). Response to any stress introduced will be recorded via change in temperature, humidity & VOC levels of the plants.

Project 6: Machine learning driven self-adaptive hardware design of electronic circuits

- Faculty: Zia Abbas
- TA: Geethika

The project is to develop a simple circuit (hardware) that can adapt by itself to the changes in voltage, temperature, etc. in order to negate all or part of their effects on the circuit performance characteristics. To do so, the circuit will be represented by a machine learning technique implemented on Arduino board. An attached AI engine will infer from the model, the input signals, the environmental conditions and create the adaptive changes required to modify the circuit's behaviour to negate all or part of their effects on the circuit performance characteristics.

Project 7: AI-Based Fatigue, Sleep Readiness, and Drowsiness Detection Using MultiSensor Wearable Data

- Faculty: Abhishek Srivastava
- TA: Karthik Gadre

Recent advances in wearable sensing technologies have enabled continuous monitoring of physiological parameters such as heart rate (HR), heart rate variability (HRV), photoplethysmography (PPG), electrodermal activity (EDA/GSR), skin temperature, respiration, and body motion. These physiological measurements, together with contextual information such as physical activity, posture, ambient lighting, and time of day, provide valuable indicators of an individual's fatigue level, sleep readiness, and drowsiness state. Machine learning techniques have demonstrated significant potential for learning complex relationships between multimodal physiological signals and human behavioural states, enabling the development of intelligent wearable systems for health monitoring, occupational safety, transportation, and human performance assessment.

The objective of this project is to develop machine learning algorithms capable of automatically estimating fatigue, sleep readiness, and drowsiness using publicly available multimodal physiological datasets. The developed framework should investigate suitable signal preprocessing techniques, physiological feature extraction methods, machine learning models, and performance evaluation metrics. Students are expected to compare multiple learning algorithms, analyse the contribution of different sensing modalities, and investigate the feasibility of lightweight models suitable for deployment on future wearable embedded platforms.

Project 8: Moving source localization using MEMS Microphones

•Faculty: Harikumar Kandath

•Mentor: Yash Mani Prakash

TA: Sai Poojith

The objective is to find the x-y coordinates of a moving sound source using MEMS microphones. Multiple Microphones will be connected to 3 laptops acting as stationary nodes. The sound signal received on these microphones will be analyzed in centralized sensor fusion architecture to estimate the accurate position of a moving source emitting a sound signal. Methods like TDOA, power based localization, or data driven methods can be used to estimate the coordinates of the moving target.

Steps involved:

- 1) Calibration of MEMS microphones
- 2) Data collection based on moving the sound source along different paths (straight line, circle, ellipse etc)
- 3) Testing and reporting accuracy of different algorithms for source localization: This needs communicating sensor data from different nodes (laptops) and develop an architecture for centralized data fusion.

Project 9: Bridge Health Monitoring

- Faculty: Sachin Chaudhari
- Mentors: Anushka, Sagar and Sampada
- TA: Anumay

- **Motivation:** Bridges are critical public lifelines; detecting degradation early protects human lives, preserves vital infrastructure, and prevents massive economic downtime.
- **The IoT Solution:** Smart sensors continuously capture structural strain, vibrations, and environmental shifts in real time, automatically triggering safety alerts and predictive maintenance before failure occurs. For example, accelerometers for vibration, strain gauges for load capacity, tiltmeters for dynamic displacement, scour sensors, and corrosion sensors.

Project 10: Multimodal Data Driven Two-Wheeler Driving Event Recognition and Quality Determination

- Faculty: Deepak Gangadharan
- TA: Geethika Palla
- Abstract: This work looks at building a hardware system that collects multimodal data (image + IMU based) about the environment in which a two-wheeler is riding. The data will be used in a synchronized manner to recognize some important driving events and determine the riding quality. The main objectives are as follows:
 - Build the hardware system with the camera and IMU sensor
 - Collect the riding data with multiple riders and label the data
 - Develop a data fusion-based event recognition algorithm for driving events and quantify the accuracy.
 - Develop an algorithm to determine the quality of riding

Project 11: VitalMind: Agentic Edge AI System for Multi-Vital Patient Monitoring - From Validated Pipeline to Clinical Deployment

- Faculty: Priyesh Shukla
- Mentor: Aryama Murthy; TA: Sai Poojith
- External mentor: Dr. Sai Tarun Vasala (Sr. Surgical Gastroenterologist – AIG Hospitals)
- Continuous multi-vital patient monitoring today relies on bulky, wired, or cloud-dependent systems that struggle in resource-constrained or remote hospital settings, delaying anomaly detection and clinician response. This project addresses that gap by building an integrated edge PCB combining a processing module, voice sensor, camera, and Bluetooth reception from a wireless multi-vital patch, enabling on-device sensor fusion and inference without constant cloud dependency. Having validated the sensor pipeline and model in Phase I, the current challenge is optimizing an agentic AI decision layer for clinical triage, collecting real-world hospital data, validating accuracy on live patients under IRB oversight, and finalizing a clinical-grade integrated PCB for deployment.

Project 12: Automated Integrated Sensing

- Faculty: A Sarje
- TA: Goutam Sutradhar (mentor 1) & Venkat Lakshamana (mentor 2)
- **Motivation:** Resistive and capacitive/inductive sensors are ubiquitous. They are used for myriad number of monitoring & data collection. The core sensor is the passive device which needs to be integrated for smart monitoring platforms
- **The IoT Solution:** Integrate R &/or C sensors [example: MQ type] to achieve automated sensing. Key points (a) adaptive biasing for the sensor for its optimal response [characterize the sensor] (b) automated digital readout with optimal adaptive sampling rate and battery monitoring (c) real time data analysis & decision making.

Project 13: Closed-Loop Stepper Control for precise angular motion

- Faculty: A Sarje
- Mentor: Goutam Sutradhar; TA: Geethika
- **Motivation:** Develop a closed-loop stepper motor system capable of achieving arc-second range angular precision and strict positional repeatability.
- **The IoT Solution:** Interface a microcontroller with a NEMA 17 stepper motor and a motor driver to achieve fine open loop angular movement (arc sec range).
- Integrate a high-resolution rotary encoder to continuously track the actual physical angle of the motor shaft.
- Program a closed-loop algorithm (like PI, PID, FOC, etc.) on the microcontroller to calculate the difference between the target angle and the actual angle for error reduction. If needed have to do noise analysis and mitigation strategies.
- Validate the final angular precision and movement repeatability by testing the motor's position against an independent, highly accurate angular measurement tool.

Project 14: Low Cost Edge-AI Vehicle Counting over LoRaWAN

- Faculty: Suresh Purini
- TA: Anumay
- Abstract: This project aims to design and evaluate a low-cost, easy-to-install roadside vehicle-counting system built around an ESP32-S3 camera node. The device will process camera images locally, count vehicles, and transmit only compact traffic summaries over LoRaWAN. Images and videos will remain on the device, reducing communication cost and improving privacy. The project will compare three approaches:
 - Classical image-processing methods
 - Lightweight neural-network inference using ESP-DL
 - A hybrid method that invokes neural inference only when motion is detected

QIDK-based projects

Project 1: AI Tutor for Remote Labs

- Faculty Guide: Sachin Chaudhari
- TA: Ansh
- Many institutions—especially in under-resourced areas—lack adequate lab infrastructure, an issue exacerbated when physical access is restricted. IoT-based Remote Triggered Labs (RTLs) bridge this gap by enabling students to conduct real-world experiments and analyze real-time data remotely via smart devices.
- This proposal enhances IoT-based remote labs by integrating QIDK edge AI kits. Featuring a local LLM-powered tutor for personalized learning the project transforms remote labs into intelligent and student-friendly environments.

Project 2: Agentic Edge AI for Smart Cities

- Guide: Anuradha Vатtem
- TA: Lokabhiram Chintada
- Existing smart building and smart city platforms typically monitor individual infrastructure domains such as air quality, energy, water, weather, and occupancy independently, relying on rule-based dashboards and threshold-based alerts. These systems lack the ability to reason across heterogeneous sensor data, identify root causes, and provide context-aware decision support. There is a need for an edge-native, privacy-preserving intelligent assistant that can autonomously coordinate specialized AI agents across multiple infrastructure domains. This work proposes an on-device multi-agent AI framework on the Qualcomm QIDK, where lightweight Small Language Models (SLMs) orchestrate domain-specific agents to analyze real-time IoT data, perform cross-domain reasoning, generate predictive insights, explain anomalies, and provide actionable recommendations through a conversational interface with visualizations and intelligent notifications.

Project 3: Edge AI for Net Zero

- Guide: Anuradha Vатtem
- TA: Lokabhiram Chintada
- Current Net Zero management systems predominantly serve as centralized monitoring platforms that collect and visualize sustainability metrics from energy, water, waste, and environmental systems. While they provide visibility into resource usage, they offer limited capability for continuous on-site analytics, real-time optimization, or localized decision-making. Their dependence on cloud infrastructure increases communication overhead, introduces delays in responding to changing conditions, and limits operation in environments with unreliable connectivity. There is a need for an edge AI-enabled sustainability platform that performs multimodal data processing, predictive modeling, and resource optimization directly on edge devices. By leveraging the Qualcomm QIDK platform, the proposed solution enables distributed intelligence for Net Zero operations through low-latency, energy-efficient, and privacy-preserving on-device AI.

Project 4: FieldMind: An On-Device Multi-Agent Copilot for Offline Field Technicians

- Faculty: Priyesh Shukla
- TA: Aryama Murthy
- Multi-agent AI systems are moving to the edge, where agents for perception, planning, tool-calling, and verification must cooperate under tight latency, power, and offline constraints. Current edge deployments accelerate model inference (NPU) but ignore the heterogeneity of agentic workflows — reasoning/chain-of-thought is compute-bound and NPU-suited, while tool calls, sensor IO, retrieval, and branching logic are CPU-bound and unpredictable. Sequential execution wastes time and battery by letting these block each other. FieldMind, an offline multi-agent field-repair copilot on QIDK, exemplifies this: technicians need real-time diagnosis via camera, sensors, and manuals with zero connectivity — demanding a scheduler that intelligently distributes tasks across NPU, CPU, and DSP.

Project 5: Exploring AI-Enabled Aerial Manipulation Using the Qualcomm QIDK Platform

- **Faculty: Spandan Roy**
- **Mentor:**
- **TA: Aryama Murthy**

- **Project Overview**

The objective of this project is to explore the capabilities of the Qualcomm QIDK platform for aerial robotics and develop a basic aerial manipulation software pipeline. Students will gain hands-on experience with embedded AI hardware, onboard perception, and robotics software by understanding the platform, interfacing with sensors, and implementing perception and manipulation algorithms suitable for aerial robotic applications.

Project 6: Multi-Label Crop Symptom Detection

Training Model and QIDK NPU Benchmarking

- Faculty: Krishna Reddy
- Mentor: Vilal Ali ; TA: Anushka
- This project focuses on the core vision & optimisation layer of the edge-based crop symptom and disease detection system. Project is responsible for model training, dataset curation, model quality (accuracy against ground truth), quantisation optimisation, and low-level NPU hardware benchmarking. You are not responsible for building a user interface or an interactive application. Your goal is to deliver an optimised, highly accurate, NPU-compatible model container with documented execution metrics on the Snapdragon QIDK hardware.

Detailed Project Objectives:

- (1) Multi-Label Dataset Curation
- (2) Model Training & Accuracy Optimization
- (3) Qualcomm Toolchain Compilation
- (4) Hardware Quantization & Quality Profiling
- (5) NPU Performance Profiling: Profile execution latency, memory bandwidth, and power consumption across QIDK CPU, GPU, and Hexagon NPU.

Project 7: Multi-Label Crop Symptom Detection Application and QIDK NPU Integration

- Faculty: Prof. Krishna Reddy
- Mentor: Vilal Ali; TA: Anushka
- This project focuses on the Systems & UI Layer of the edge-based Crop Symptom and Disease Detection system. Project is responsible for application engineering, system integration, local database design, and the interactive flow. Your grading is based on software execution and functionality (making the end-to-end pipeline run smoothly offline on the QIDK kit without crashing). You are not expected to train machine learning models or optimize their accuracy; you will deploy a pre-existing/standard public model on the NPU and connect its outputs to your React (React Native / React Web) application.

Detailed Project Objectives

1. Offline React Development
2. QIDK NPU SDK Integration
3. Interactive Questionnaire Engine
4. Local Disease Mapping
5. On-Device Offline Execution

Project 8: Trust-aware Federated Learning Under Hardware-induced Distribution Shift

- Faculty: Saikiran Bulusu
 - TA: Alan/Lokabhiram Chintada
- Project Overview
 - **Edge AI devices** operate under varying hardware and environmental conditions, leading to **hardware-induced distribution shifts**.
 - **Conventional federated learning** treats all client updates equally, making the global model susceptible to unreliable or drifted devices.
 - **Proposed approach:** A **trust-aware federated learning framework** that uses QIDK telemetry (temperature, power, latency, utilization) to estimate client reliability.
 - **Trust-weighted aggregation** prioritizes reliable client updates, improving the robustness and long-term reliability of privacy-preserving edge AI systems.

Project 9: Radio Frequency mapping using QIDK

- Faculty: Arti Yardi and Sachin Chaudhari
- External Collaborator: Ramana, MathWorks
- Mentor: Adeeb; TA: Vedant
- Project Overview
 - **Radio frequency (RF) mapping** corresponds to creating a heatmap of the radio signal behavior of within a given area. In this project, the goal is to create RF map of the given room using the existing popular algorithms
 - **Objective-1:** Create 3D model of the room using 2D images
 - **Objective-2:** Run two popular RF mapping algorithms on QIDK (which are available in MATLAB libraries)
 - **Objective-3:** Benchmark all the results against LIDAR (for Objective-1) and against MATLAB (for Objective-2)

Project 10: Evaluation of image inpainting algorithms and their optimization on QIDK platform

- Faculty: Deepak G.
- TA: Vedant Tejas
- Project Overview:

Image Inpainting has become a widely used application in mobile devices. It uses the surrounding pixel information and AI algorithms to reconstruct missing parts of an image. The state-of-the-art algorithms use GAN models to perform the task. The objectives of the project will be

 - Investigate the models in literature for Image Inpainting
 - Choose few models and implement them on the QIDK platform
 - Conduct performance optimization for these models

Project 11: Non-Contact Vitals Monitoring Using QIDK

- Faculty: Abhishek Srivastava
- TA: Alan
- **Millimetre-wave (mmWave) radar** operates in the 30–300 GHz frequency range and uses Frequency-Modulated Continuous-Wave (FMCW) waveforms to estimate target range, velocity, and angle. Radar processing typically includes range, Doppler, and angle estimation, producing representations such as point clouds, range-Doppler maps, and micro-Doppler signatures for applications including detection, tracking, classification, and behaviour analysis. In non-contact vital signs monitoring, mmWave radar measures minute chest movements caused by respiration and cardiac activity by analysing phase variations in the reflected FMCW signals. The processing pipeline generally includes target localisation, clutter suppression, phase extraction, filtering, and spectral analysis to estimate physiological parameters such as respiration rate and heart rate without requiring physical contact.

Project 12 : Evaluation of AQI Estimation Algorithms

- Faculty Guide: Sachin Chaudhari
- TA: Ansh
- Context: There are several issues in sensor based air pollution monitoring: failures, maintenance, scalability. To overcome that, we have designed camera based air pollution estimation algorithms.
- Project Overview:
 - Implement and Compare **Transformer-based and CNN-based AQI Estimation Frameworks using** on QIDK
 - **Compare** model performance, metrics, efficiency, and robustness across multiple test cases.
 - **Evaluate** the two frameworks and analyze the trade-offs between them
 - **Benchmark** the results on provided dataset
 - Evaluation can include **ablation** as well as deployment-oriented analysis, including **practical challenges**.

Course Logistics

Project Distribution

- Fill in the form and preferences by 05 Aug. 2026
- First preferences, then lottery
- Projects allotted by 07 Aug. 2026 by Head TAs

Course Portal

- Moodle
 - Notifications
 - News
 - Discussion Forum
 - Submissions

Evaluation Plan

- Mark distributions (tentative): 100 for project
 - Evaluation for labs (Aug): 20 (by respective faculty/mentors)
 - Evaluation for labs (Sep): 20 (by Sachin+Aftab+Deepak)
 - Evaluations for labs (Oct): 20 (by respective faculty)
 - Overall: 40 (Externals + Deepak+Aftab+Sachin+Internal Faculty/Qualcomm team)
- Grading: Relative (TBD)

Attendance Policy

- Labs
 - Once per week on Thursday 2-5 pm
 - Mandatory
 - Attendance will be taken 2-3 times
- Classes
 - Only occasionally, but mandatory attendance.

Reimbursement for Project Sensors

- **Most of the components are expected to be in the lab**
- Students will be reimbursed for the project-specific sensors and other electronic components which needs to be additionally procured
 - Please consult your faculty, TA and Lab Manager (Mr. Prashant Ranjan) before purchasing the project-specific sensors
 - Get a written approval from respective faculty over email cc:ing Lab Manager (Mr. Prashant Ranjan) and TA
- Maximum up to Rs. 5 K per team!
 - In case going above it, need to request the faculty coordinators with requested budget with justification.
 - You can always buy sensors on your own and add to the project
- Bills should be proper GST bills (not quotations/estimates) and in the name of IIIT Hyderabad.
- Note that all the reimbursement bills must be submitted all components should be returned by 8 Dec. 2026.

Questions?

- That's all for today!
- See in you in lab on 6 Aug. 2026